Boyang Zhou

Ph.D. Student, Wireless@Virginia Tech

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Summary

Fifth-year Computer Engineering Ph.D. student at the Wireless@Virginia Tech Lab. Specializing in wireless communications across the RF, PHY, and MAC domains. Applies communication theory, mathematical optimization, and machine learning to the design and optimization of reliable, resource-efficient wireless systems. Expertise spans RF impairment modeling, link-level performance analysis, digital signal processing (DSP) algorithm design, and MAC-layer resource allocation and scheduling.

Education

Ph.D. in Computer Engineering, Virginia Tech — Aug 2022 — (Expected) May 2027
Advisor: Y. Thomas Hou

M.S. in Computer Engineering, Virginia Tech — Aug 2022 — Dec 2024
GPA: 3.9/4.0

B.S., The Pennsylvania State University — Aug 2018 — May 2022
Dual Degree in Computer Science and Mathematics
GPA: 3.91/4.00, *Magna Cum Laude*

Work Experience

Graduate Research Assistant — Aug 2022 — Present
Wireless@Virginia Tech

- Performed RF-aware wireless system analysis by accounting for RF impairments and transceiver nonidealities and characterizing their effects on baseband signal processing, multiuser interference, effective SINR, and spectrum utilization.
- Designed and implemented a multiuser MIMO-OFDM link-level simulation covering 3GPP channel models, RF and PHY nonidealities, channel estimation, multiuser beamforming, modulation, and channel coding and decoding.
- Developed millisecond-scale beamforming and MAC-layer scheduling algorithms under practical RF- and PHY-layer constraints to satisfy stringent latency and reliability requirements.
- Utilized communication theory, real-time GPU algorithm design, convex optimization, machine learning, and reinforcement learning to address these problems under tight running time constraints of under 5 milliseconds.

Projects

MIMO-OFDM Link-Level Simulator

- Developed a MATLAB link-level simulation framework for MIMO-OFDM system design and performance evaluation.
- Modeled AWGN and time- and frequency-selective multipath channels with Rayleigh, Rician, and Nakagami-(m) fading, including delay spread, Doppler spread, and spatial antenna correlation.
- Implemented OFDM modulation and demodulation, cyclic-prefix insertion and removal, pilot-based LS channel estimation, channel equalization, error-correction coding, interleaving, and adaptive subcarrier bit loading.
- Implemented OFDM carrier-frequency and symbol-timing synchronization to estimate and compensate for carrier-frequency and timing offsets.
- Implemented MIMO transmission and reception techniques, including MRT transmit beamforming, MRC receive combining, SVD-based spatial multiplexing, Alamouti space-time block coding, and ZF, MMSE, SIC, and Maximum Likelihood detection.
- Evaluated BER, BLER, throughput, and spectral efficiency and quantified the trade-off between pilot overhead and transmission performance.

End-to-End RAN Prototype Using MIMO (Team Project)

- Designed and implemented a QoS-aware MAC-layer scheduler for an end-to-end 2×2 MIMO RAN prototype for next-generation cellular networks.
- Designed the scheduler to allocate radio resources based on application throughput and latency requirements and real-time feedback from the RAN.
- Evaluated the scheduler against throughput, latency, and packet loss on the end-to-end prototype.

End-to-End Software-Defined Cellular Network Deployment and Control

- Deployed an end-to-end software-defined cellular system using the NSF Open AI Cellular (OAIC) platform and verified connectivity across the core network, base station, user equipment, and virtual radio link.
- Configured an Open Radio Access Network (O-RAN) near-real-time RAN Intelligent Controller (near-RT RIC) and connected it to an **srsRAN** base station for network monitoring and control.

- Conducted hands-on experiments with near-real-time RAN control applications (xApps) for key performance indicator (KPI) monitoring, radio-resource scheduling, and AI-driven network control using OAIC's controller and automated testing frameworks.

Publications

Conference paper

- **B. Zhou** (First author) Coauthors and title redacted for double-blind review
  Submitted for *IEEE INFOCOM 2027*
- **B. Zhou** (Second author) Coauthors and title redacted for double-blind review
  Submitted for *IEEE INFOCOM 2027*

Journal paper

- **B. Zhou**, Y. Shi, Y. Wu, S. Acharya, L. DaSilva, W. Lou, and Y. T. Hou (2026),“ Atalanta: A Bandwidth-Conserving URLLC Scheduler for Industrial Automation,” under major revision for *IEEE Internet of Things Journal*.
- Q. Liu, C. Li, S. Li, Y. Shi, **B. Zhou**, Y. T. Hou, and W. Lou, “Pistis: A Scheduler to Achieve Ultra Reliability for URLLC Traffic in 5G O-RAN,” *IEEE Internet of Things Journal*, vol. 11, no. 21, pp. 35405–35419, November 2024.
- J. Zhao, **B. Zhou**, J. P. Butler, R. G. Bock, J. P. Portelli, and S. G. Bilén, “IoT-based sanitizer station network: A facilities management case study on monitoring hand sanitizer dispenser usage,” *Smart Cities*, vol. vol. 4, no. 3, pp.979–994, July 2021.

Video Demo

- CNSR@VT, “A Demo of Data Scheduling for Ultra-Reliable Low Latency Communications in O-RAN,” *Video*, YouTube, 2024. [Online]. Available: https://www.youtube.com/watch?v=LNp8X76iEpA. [Accessed: Sep. 3, 2026].

Selected Honors and Awards

First Place, *Open6G+AI Workshop Hackathon* Apr. 2026

- Won first place in a team competition with participants from more than 30 institutions worldwide
- Helped design the MAC-layer scheduler around application QoS requirements and RAN feedback; Contributed to its implementation and integration, and diagnosed deployment and connectivity failures during end-to-end testing.
- Competition organized by Northeastern University's Institute for Intelligent Networked Systems.

Cyber Innovation Scholar, *Commonwealth Cyber Initiative Southwest Virginia* Apr. 2026

- Selected to explore the practical impact and commercialization potential of cyber and wireless research.
- Received $1,500 in professional-development funding for conference travel and technical training.
- Completed research-commercialization and entrepreneurship training through a cyber startup workshop led by Virginia Tech LAUNCH.

Department Research Fellowship, *Virginia Tech ECE* Aug. 2022

- Selected as the sole fellowship recipient among the department's incoming Ph.D. students.
- Received approximately $70,000 in funding covering full tuition and a stipend for the first year of the Ph.D. program.

First Place, *CodePSU Programming Competition, Beginner Division* Mar. 2019

- Won the beginner division of a campus-wide algorithmic programming competition.
- Competition organized by Penn State's student chapter of the Association for Computing Machinery.

Other Experiences

Math Guided Study Group Leader March 2020 — May 2022
The Pennsylvania State University

- Review the knowledge students learned during the past week
- Communicate with professors and students to go through students' questions

Math Tutor Sept 2019 — May 2022
The Pennsylvania State University

- Weekly assist more than 15 students one-on-one in math classes up to and including Calculus II and below

Skills

Programming:	C, CUDA, C++, Python, Pytorch, Matlab, Verilog, Java, SQL, Git
Software:	Adobe Photoshop, After Effects, Premiere
Languages:	English, Chinese, French (B1 level)